

Chapter: 02

There might be
some mistakes in
the calculation

Practice Problems: Chapter 2

1. Evaluate the interpolation polynomial of a function $f(x) = \sin(x)$ at the nodes $\{-\pi/2, 0, \pi/2\}$ by using the Vandermonde method. Note you need to show a detailed calculation to get full credit.
2. Consider the function $f(x) = xe^x$. Using Taylor Expansion a polynomial of degree 3 is calculated.
 - a. Using Taylor Expansion, write $f(x)$ as an infinite series.
 - b. Find the coefficients a_0, a_1, a_2 and a_3 for the polynomial $P_3(X)$.
 - c. Compute $f(0.1)$ and $P(0.1)$ up to seven significant figures.
 - d. Find the percentage of error using the answers got from Question (c)
3. Consider the function $f(x) = \ln(1 + x)$. Using Taylor Expansion a polynomial of degree 3 is calculated.
 - a. Using Taylor Expansion, write $f(x)$ as an infinite series.
 - b. Find the coefficients a_0, a_1 and a_2 for the polynomial $P_2(X)$.
 - c. Compute $f(0.2)$ and $P(0.2)$ up to seven significant figures.
 - d. Find the percentage of error using the answers got from Question (c)
4. A function, $f(x) = x \sin(x)$, is to be interpolated at three nodes: $x_0 = -\pi/2, x_1 = 0$ and $x_2 = \pi/2$. Answer the following:
 - a. Draw a sketch of the function $f(x)$ and identify the nodes. Briefly explain what kind of interpolation polynomial is expected and discuss the properties of the polynomial.
 - b. Compute Lagrange bases.
 - c. Write the interpolation polynomial and compare the properties of the interpolation polynomial with the expectation in Part-(a).
5. Read the following and answer accordingly:
 - a. Find an interpolating polynomial of appropriate degree using Newton's divided-difference method for $f(x) = x \cos(x)$. Consider the nodes $[-\pi/2, 0, \pi/2]$.
 - b. Use the interpolated polynomial to find an approximate value at $\pi/4$, and compute the relative error at $\pi/4$.
 - c. Add a new node π to the above nodes, and find the interpolating polynomial.
6. Given, $f(x) = \sin(x)$ and Consider the nodes $[0, \pi/2, \pi]$. Use Newton's Divided-Difference method for the following questions-
 - a. Find the values of a_0, a_1 and a_2 .
 - b. Write down the interpolating polynomial.
 - c. Add a new node $3\pi/2$ to the existing nodes and find a new interpolating polynomial.
 - d. Estimate the upper bound of the interpolating error for the polynomial $P_3(x)$.

7. Consider the following table of data points/nodal points:

Time (sec) t	Velocity (ms^{-1}) $v(t)$
2	10
4	20
6	25

- Find an interpolating polynomial of velocity that goes through the above data points by using Vandermonde Matrix method. Also compute an approximate value of acceleration at Time, $t = 7$ sec.
 - Find an interpolating polynomial of velocity that goes through the above data points by using Lagrange method.
 - If a new data point is added in the above scenario, which method should you use in finding a new interpolating polynomial? Also what will be the degree of that new polynomial?
8. Use a Hermite polynomial that agrees with the data listed below to find the approximation of $f(1.5)$.

k	x_k	$f(x_k)$	$f'(x_k)$
0	1.3	0.6200860	-0.5220232
1	1.6	0.4554022	-0.5698959
2	1.9	0.2818186	-0.5811571

9. Find the value of y for $x = 2.1$ using 2nd order Lagrange polynomials with the appropriate data.

SI.	x	y
1	-1	2.2
2	0	10.6
3	1	17.0
4	2	22.4
5	3	25.8

10. Let $f(x) = \cos(x)$ and the nodes are $(-\pi/4, 0, \pi/4)$. Use Cauchy's theorem to find the upper bound of the error.

11. a. Compute the upper bound of error using Cauchy's Theorem for $f(x) = \sin^2(x/2)$ with nodes $\{-\pi/3, 0, \pi/3\}$ within the interval $[-1.2, 1.2]$.
 b. Why are Chebyshev nodes an optimal choice in Interpolation?

12. The following nodes come from the function $f(x) = \ln(5x+9)$:

x	f(x)
-0.5	1.87
0	2.20
0.5	2.44

- a. Using Newton's divided difference method, find the equation of a second degree polynomial which fits the above data points.
 b. Expand the function $f(x) = \ln(5x + 9)$ using Taylor Series, centered at 0. Include till the x^2 term of the Taylor series.
 c. Should the equation which you found in part (a) and part (b) match? Comment on why, or why not.

13. Consider the following nodes:

x	f(x)
0	5
3	9.5
6	5

- a. If an equation of a polynomial which fits through the above nodes is found using both the Vandermonde Matrix approach and the Lagrange approach, will both the equations match?
 b. Find the equation of a polynomial which fits through the above nodes using the Vandermonde matrix approach.
 c. Find the equation of a polynomial which fits through the above nodes using the Lagrange approach.

14. Given a Runge Function, $f(x) = \frac{8}{5+16x^2}$ between the interval $[-6,6]$ where $n=3$

- a. Calculate the equal angled points of θ 's.
 b. Calculate the values of the Chebyshev's nodes.
 c. Find the Lagrange basis $l_2(x)$.

15. Consider the following dataset:

x	$f(x)$	$f'(x)$
0.1	-0.62050	3.58502
0.2	-0.28340	3.14033

Answer the following based on the above data:

- Compute the Hermite bases: $h_0(x)$, $h_1(x)$, $\hat{h}_0(x)$ and $\hat{h}_1(x)$.
- Write the Hermite polynomial and find the value at $x = 0.15$

Practice Sheet-02

Chapter-02

Date: / /

3

x	$f(x) = \sin x$
$-\pi/2$	-1
0	0
$\pi/2$	1

nodes = 3

degree = 3-1 = 2

$$P_2(x) = a_0 x^0 + a_1 x^1 + a_2 x^2$$

$$P_2(-\pi/2) = a_0 + a_1(-\pi/2) + a_2(\pi/2)^2 = -1$$

$$P_2(0) = a_0 + a_1(0) + a_2(0)^2 = 0$$

$$P_2(\pi/2) = a_0 + a_1(\pi/2) + a_2(\pi/2)^2 = 1$$

$$\begin{bmatrix} 1 & -\pi/2 & (\pi/2)^2 \\ 1 & 0 & 0 \\ 1 & \pi/2 & (\pi/2)^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & -\pi/2 & (\pi/2)^2 \\ 1 & 0 & 0 \\ 1 & \pi/2 & (\pi/2)^2 \end{bmatrix}^{-1} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.6366 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore P_2(x) =$$

$$0.6366x$$

4 (a)

x	$f(x) = x \sin x$
$-\pi/2$	1.5707
0	0
$\pi/2$	1.5707

$$f(x) = xe^x$$

Date: / /

(2) (a) $P(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3$

(b) If x_0 value not given then consider $x_0 = 0$.

$$a_0 = f(x_0) = 0$$

$$a_n = \frac{f^n(x_0)}{n!}$$

$$a_1 = f'(x_0) = x_0 e^{x_0} + e^{x_0}, f'(0) = 1$$

$$a_2 = \frac{f''(x_0)}{2!} = \frac{x_0 e^{x_0} + e^{x_0} + e^{x_0}}{2!}, f''(0) = \frac{2}{2!} = 1$$

$$a_3 = \frac{f'''(x_0)}{3!} = \frac{1}{3!} (x_0 e^{x_0} + e^{x_0} + e^{x_0} + e^{x_0}), f'''(0) = \frac{3}{3!} = \frac{3}{6} = \frac{1}{2}$$

$$a_0 = 0, a_1 = 1, a_2 = 1, a_3 = \frac{1}{2}$$

$$\underline{=} f(0.1) = (0.1)e^{(0.1)} = 0.1105170$$

$$P_3(0.1) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3$$

$$= 0 + 1 \times (0.1 - 0) + 1 \times (0.1 - 0)^2 + \frac{1}{2} (0.1 - 0)^3$$

$$= 0.1 + (0.1)^2 + (0.1)^3 \times \frac{1}{2} = 0.1105000 \text{ [upto 7 significant digit]}$$

$$\underline{=} \text{Accurate} = 0.1105170, \text{Approx} = 0.1105000$$

$$\text{Percentage Error} = \left| \frac{0.1105170 - 0.1105000}{0.1105170} \right|$$

$$= 0.01538\%$$

(2) (a) $P(x) = x_0 e^{x_0} + (x-x_0) \cdot (x_0 e^{x_0} + e^{x_0}) + \frac{(x-x_0)^2}{2!} (x_0 e^{x_0} + 2e^{x_0}) + \frac{(x-x_0)^3}{3!} (x_0 e^{x_0} + 3e^{x_0})$

$$f(x) = \ln(1+x)$$

Date: / /

(3)

$$a \quad p(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3$$

$$b \quad a_0 = f(x_0) = \ln(1+0) = \ln 1 = 0$$

let, $a_0 = 0$

$$a_1 = f'(x_0) = \frac{1}{1+x} = \frac{1}{1+0} = 1$$

$$a_2 = \frac{f''(x_0)}{2!} = \frac{-1(1) + (1+x) \cdot 0}{(1+x)^2} = \frac{-1}{2(1+x)^2} = \frac{1}{2}$$

$$a_0 = 0$$

$$a_1 = 1$$

$$a_2 = 1$$

$$c \quad f(0.2) = 0.1823215$$

$$p(x) = 0 + f(x)$$

$$p(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2$$

$$p(0.2) = 0 + 1(0.2-0) + \frac{1}{2}(0.2-0)^2$$

$$= 0.1800000 \quad [\text{upto 7 significant digit}]$$

$$d \quad \text{Actual} = 0.1823215, \quad \text{Approx} = 0.1800000$$

$$\therefore \text{error percentage} = \left| \frac{0.1823215 - 0.1800000}{0.1823215} \right| \times 100\%$$

$$= 1.2733\%$$

0	0
1	$\frac{1}{\sqrt{\pi}}$

$$x_0 \cdot 0 + x_1 \cdot 0 + x_2 \cdot 0 = (x)_0$$

$$1 = -\frac{1}{\sqrt{\pi}}(x) \cdot 0 + (x - \frac{1}{\sqrt{\pi}}) \cdot 0 + 0 = (x - \frac{1}{\sqrt{\pi}})_0$$

$$0 = 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 = (0)_0$$

4. (b) $P_2(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2) \quad \text{--- (i)}$

$$l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-0)(x-\frac{1}{\sqrt{\pi}})}{4 \cdot 0.34} \quad \boxed{P_n(x) = \sum_{k=0}^n l_k(x)f(x_k)}$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x+\frac{1}{\sqrt{\pi}})(x-\frac{1}{\sqrt{\pi}})}{(0+\frac{1}{\sqrt{\pi}})(0-\frac{1}{\sqrt{\pi}})} = \frac{(x+\frac{1}{\sqrt{\pi}})(x-\frac{1}{\sqrt{\pi}})}{-(\frac{1}{\sqrt{\pi}})^2}$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x+\frac{1}{\sqrt{\pi}})(x-0)}{(\frac{1}{\sqrt{\pi}}+\frac{1}{\sqrt{\pi}})(\frac{1}{\sqrt{\pi}}-0)} = \frac{x(x+\frac{1}{\sqrt{\pi}})}{\frac{(\frac{1}{\sqrt{\pi}})^2}{2}}$$

4. c $P_2(x) = \left\{ \frac{2x(x-\frac{1}{\sqrt{\pi}})}{(\frac{1}{\sqrt{\pi}})^2} \times 1.5707 \right\} + \left\{ \frac{x^2 - (\frac{1}{\sqrt{\pi}})^2}{-(\frac{1}{\sqrt{\pi}})^2} \times 0 \right\} + \left\{ \frac{x(x+\frac{1}{\sqrt{\pi}}) \cdot 2}{(\frac{1}{\sqrt{\pi}})^2} \times 1.5707 \right\}$

$$= 0.15914 (2x^2 - \pi x) + 0.15914 (2x^2 + \pi x)$$

$$= 0.15914 \times 4x^2 = 0.63656x^2 = \frac{2}{\pi} x^2$$

$$x_0 = -\pi/2, f[x_0] = 0$$

$$x_1 = 0, f[x_1] = 0$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = 0$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = 0$$

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2} = 0$$

new node:

$$x_3 = \pi, f[x_3] = -\pi$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = 0$$

$$f[x_1, x_2, x_3] = \frac{f[x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{-\pi - 0}{\pi - 0} = -1$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{-1 - 0}{\pi - (-\pi/2)} = -\frac{2}{3\pi/2} = -\frac{4}{3\pi}$$

$$f[x_0, x_1, x_2, x_3, x_4] = \frac{f[x_1, x_2, x_3, x_4] - f[x_0, x_1, x_2, x_3]}{x_4 - x_0} = \frac{0 - (-\frac{4}{3\pi})}{\pi - (-\pi/2)} = \frac{4}{3\pi/2} = \frac{8}{3\pi}$$

$$f[x_1, x_2, x_3, x_4] = \frac{f[x_4] - f[x_1, x_2, x_3]}{x_4 - x_1} = \frac{0 - (-\frac{4}{3\pi})}{\pi - 0} = \frac{4}{3\pi}$$

$$f[x_2, x_3, x_4] = \frac{f[x_4] - f[x_2, x_3]}{x_4 - x_2} = \frac{0 - 0}{\pi - \pi/2} = 0$$

$$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{x_4 - x_3} = \frac{0 - (-\pi)}{\pi - \pi} = \text{undefined}$$

$$f[x_4] = 0$$

(5) (6)

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = 0$$

$$f[x_1, x_2, x_3] = \frac{f[x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{-2 - 0}{\pi - 0} = -\frac{2}{\pi}$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{-\frac{2}{\pi} - 0}{\pi - (-\pi/2)} = -\frac{2/\pi}{3\pi/2} = -\frac{4}{3\pi}$$

$$f[x_1, x_2, x_3, x_4] = \frac{f[x_4] - f[x_1, x_2, x_3]}{x_4 - x_1} = \frac{0 - (-\frac{4}{3\pi})}{\pi - 0} = \frac{4}{3\pi}$$

$$f[x_2, x_3, x_4] = \frac{f[x_4] - f[x_2, x_3]}{x_4 - x_2} = \frac{0 - 0}{\pi - \pi/2} = 0$$

$$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{x_4 - x_3} = \frac{0 - (-\pi)}{\pi - \pi} = \text{undefined}$$

$$f[x_4] = 0$$

$$f[x_0, x_1, x_2, x_3, x_4] = \frac{f[x_1, x_2, x_3, x_4] - f[x_0, x_1, x_2, x_3]}{x_4 - x_0} = \frac{\frac{4}{3\pi} - (-\frac{4}{3\pi})}{\pi - (-\pi/2)} = \frac{8/3\pi}{3\pi/2} = \frac{16}{9\pi}$$

$$f[x_1, x_2, x_3, x_4, x_5] = \frac{f[x_5] - f[x_1, x_2, x_3, x_4]}{x_5 - x_1} = \frac{0 - \frac{16}{9\pi}}{\pi - 0} = -\frac{16}{9\pi}$$

$$f[x_2, x_3, x_4, x_5] = \frac{f[x_5] - f[x_2, x_3, x_4]}{x_5 - x_2} = \frac{0 - 0}{\pi - \pi/2} = 0$$

$$f[x_3, x_4, x_5] = \frac{f[x_5] - f[x_3, x_4]}{x_5 - x_3} = \frac{0 - (-\pi)}{\pi - \pi} = \text{undefined}$$

$$f[x_4, x_5] = \frac{f[x_5] - f[x_4]}{x_5 - x_4} = \frac{0 - 0}{\pi - \pi} = \text{undefined}$$

$$f[x_5] = 0$$

x	f(x)
-π/2	0
0	0
π/2	0

$$f(x) = x \cos x$$

Date: / /

$$= -\frac{2}{3\pi/2}$$

$$\leq p_3(x) = 0 + (x - x_0)(x - x_1)(x - x_2) \cdot f[x_0, x_1, x_2, x_3]$$

$$= 0 + (x + \pi/2)(x - 0)(x - \pi/2) \cdot (-\frac{4}{3\pi/2})$$

$$= -\frac{4}{3\pi/2} (x + \pi/2)(x - \pi/2) \cdot x$$

$$p_3(x) = -\frac{4x}{3\pi/2} (x^2 - (\pi/2)^2)$$

$$\therefore \text{relative error} = \frac{0.555 - 0}{0.555} \times 100\%$$

$$= 100\%$$

$$f(u) = \sin u$$

u	$f(u)$
0	0
$\pi/2$	1
π	0

$$u_0 = 0, f[u_0] = 0$$

$$u_1 = \pi/2, f[u_1] = 1$$

$$u_2 = \pi, f[u_2] = 0$$

$$f[u_0, u_1] = \frac{f[u_1] - f[u_0]}{(u_1 - u_0)}$$

$$= \frac{1 - 0}{\pi/2 - 0} = \frac{2}{\pi}$$

$$f[u_1, u_2] = \frac{f[u_2] - f[u_1]}{u_2 - u_1}$$

$$= \frac{0 - 1}{\pi - \pi/2} = -\frac{1}{\pi/2} = -\frac{2}{\pi}$$

$$f[u_2, u_3] = \frac{f[u_3] - f[u_2]}{u_3 - u_2} = \frac{-1 - 0}{3\pi/2 - \pi} = -\frac{1}{\pi/2} = -\frac{2}{\pi}$$

$$(b) P_2(u) = u_0(u) a_0 + u_1(u) a_1 + u_2(u) a_2$$

$$= f[u_0] + (u - u_0) \cdot f[u_0, u_1] + (u - u_0)(u - u_1) f[u_0, u_1, u_2]$$

$$= 0 + (u - 0) \cdot \frac{2}{\pi} + (u - 0)(u - \pi/2) \cdot (-\frac{4}{\pi^2})$$

$$= \frac{2}{\pi} u - \frac{4}{\pi^2} u(u - \pi/2) = \frac{2}{\pi} u - \frac{4}{\pi^2} (u^2 - \frac{\pi u}{2}) = \frac{2}{\pi} u - \frac{4u^2}{\pi^2} + \frac{4}{\pi^2} \times \frac{\pi u}{2}$$

$$= \frac{2}{\pi} u - \frac{4u^2}{\pi^2} + \frac{2u}{\pi} = \frac{4u}{\pi} - \frac{4u^2}{\pi^2}$$

$$\therefore P_3(u) = \frac{2}{\pi} u - \frac{4}{\pi^2} u(u - \pi/2) + (u - u_0)(u - u_1)(u - u_2) \cdot f[u_0, u_1, u_2, u_3]$$

$$= \frac{2}{\pi} u - \frac{4}{\pi^2} (u^2 - \pi/2 u) + (u)(u - \pi/2)(u - \pi) \times \frac{8}{3\pi^3}$$

$$a_0 = \frac{2}{\pi},$$

$$a_1 = 0$$

$$a_2 = -\frac{4}{\pi^2}$$

$$f[u_0, u_1, u_2] = \frac{-\frac{2}{\pi} - \frac{2}{\pi}}{\pi - 0} = -\frac{4}{\pi^2}$$

$$= -\frac{4}{\pi^2}$$

$$f[u_1, u_2, u_3] = \frac{-\frac{2}{\pi} + \frac{2}{\pi}}{3\pi/2 - \pi/2} = 0$$

$$f[u_0, u_1, u_2, u_3] = \frac{0 - \frac{4}{\pi^2}}{3\pi/2 - 0} = -\frac{8}{3\pi^3}$$

$$= -\frac{8}{3\pi^3} = -0.086004$$

Date: / /

$[0, \frac{3\pi}{2}] \rightarrow f$ range

0.04167

$$(c) (d) |f(x) - P_3(x)| \leq \left| \frac{f^{(4)}(\xi)}{4!} \right|_{\max} |(x-x_0)(x-x_1)(x-x_2)(x-x_3)|_{\max}$$

$$\leq \left| \frac{\sin(\frac{\pi}{2})}{4!} \right|_{\max} |(x-0)(x-\pi/2)$$

$$(x-\pi)(x-\frac{3\pi}{2})|_{\max}$$

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$f'''(x) = -\cos x$$

$$f^{(4)}(x) = \sin x$$

$$\leq \left| \frac{\sin(\frac{\pi}{2})}{4!} \right| \left| \left(x^2 - \frac{\pi x}{2} \right) (x-\pi) \left(x-\frac{3\pi}{2} \right) \right|_{\max}$$

$$\leq \left(\frac{1}{4 \times 3 \times 2 \times 1} \right) \left| \left(x^3 - \frac{\pi x^2}{2} - \frac{\pi x^2}{2} + \frac{\pi^2 x}{2} \right) \left(x - \frac{3\pi}{2} \right) \right|_{\max}$$

$$\leq 0.04167 \left| x^4 - \pi x^3 - \frac{\pi x^3}{2} + \frac{\pi^2 x^2}{2} - \left(\frac{3\pi x^3}{2} - \frac{3\pi^2 x^2}{2} - \frac{3\pi^2 x^2}{2} + \frac{3\pi^3 x}{4} \right) \right|_{\max}$$

$$\leq 0.04167 \left| x^4 - x^3 \left(\pi + \frac{\pi}{2} + \frac{3\pi}{2} \right) + x^2 \left(\frac{\pi^2}{2} + \frac{3\pi^2}{2} + \frac{3\pi^2}{4} \right) - \frac{3\pi^3 x}{4} \right|_{\max}$$

$$\leq 0.04167 \left| x^4 - x^3 \cdot 3\pi + \frac{11}{4} \pi^2 x^2 - \frac{3\pi^3 x}{4} \right|_{\max}$$

$$\leq 0.04167 \left| x^4 + \frac{11}{4} \pi^2 x^2 - \frac{3\pi^3}{4} x \right|_{\max}$$

$$w(x) = x^4 + \frac{11}{4} \pi^2 x^2 - \frac{3\pi^3}{4} x - 3\pi x^3$$

$$w'(x) = 4x^3 + \frac{22}{4} \pi^2 x - \frac{3\pi^3}{4} - 9\pi x^2$$

$$0 = 4x^3 + \frac{22}{4} \pi^2 x - \frac{3\pi^3}{4} - 9\pi x^2$$

$$x_1 = -54.2907 \quad 0.599999$$

$$x_2 = 4.11239$$

$$x_3 = 2.35619$$

x	$w(x)$
0	0
$\frac{3\pi}{2}$	0
0.59999	-6.08806
4.11239	-6.08806
2.35619	3.41302

Now,

$$|f(x) - P_3(x)| \leq 0.04167x \left| \frac{3.41302}{-6.08806} \right|$$

$$\leq 0.04167x \left(\frac{3.41302}{6.08806} \right)$$

$$\leq 0.1422$$

you must take the largest absolute value over the interval

$[0, \frac{3\pi}{2}]$, here $|-6.08806|$

cannot be used because that is outside the range.

7 (a) 3 nodes $\rightarrow$ 2 degree

$$v(t) = a_0 + a_1 t + a_2 t^2$$

$$v(2) = a_0 + a_1 2 + a_2 4 = 10$$

$$v(4) = a_0 + a_1 4 + a_2 16 = 20$$

$$v(6) = a_0 + a_1 6 + a_2 36 = 25$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 6 & 36 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 6 & 36 \end{bmatrix}^{-1} \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix} = \begin{bmatrix} -5 \\ 35/4 \\ -5/8 \end{bmatrix}$$

$$\therefore V_2(t) = -5 + 35/4 t - 5/8 (t)^2$$

$$a_2(t) = \frac{d}{dt}(V_2(t)) = \frac{35}{4} - \frac{5}{4} t$$

$$a_2(7) = \frac{35}{4} - \frac{5}{4} \times 7 \\ = 0 \text{ m/s}^2$$

$$t_0 = 2, t_1 = 4, t_2 = 6$$

Date: / /

$$7(b) \quad p(t) = l_0(t)v(t_0) + l_1(t)v(t_1) + l_2(t)v(t_2) \quad \text{--- (1)}$$

$$l_0(t) = \frac{(t-t_1)(t-t_2)}{(t_0-t_1)(t_0-t_2)} = \frac{(t-4)(t-6)}{(2-4)(2-6)} = \frac{1}{8} (t-4)(t-6)$$

$$l_1(t) = \frac{(t-t_0)(t-t_2)}{(t_1-t_0)(t_1-t_2)} = \frac{(t-2)(t-6)}{(4-2)(4-6)} = -\frac{1}{4} (t-2)(t-6)$$

$$l_2(t) = \frac{(t-t_0)(t-t_1)}{(t_2-t_0)(t_2-t_1)} = \frac{(t-2)(t-4)}{(6-2)(6-4)} = \frac{1}{8} (t-2)(t-4)$$

Now,

$$\begin{aligned} p(t) &= \left\{ \frac{1}{8} (t-4)(t-6) \times 10 \right\} + \left\{ -\frac{1}{4} (t-2)(t-6) \times 20 \right\} + \left\{ \frac{1}{8} (t-2)(t-4) \times 25 \right\} \\ &= \frac{5}{4} (t^2 - 6t - 4t + 24) - \left\{ 5 (t^2 - 2t - 6t + 12) \right\} + \left\{ \frac{25}{8} (t^2 - 2t - 4t + 8) \right\} \\ &= \frac{5}{4} (t^2 - 10t + 24) - 5 (t^2 - 8t + 12) + \frac{25}{8} (t^2 - 6t + 8) \\ &= t^2 \left(\frac{5}{4} - 5 + \frac{25}{8} \right) + t \left(-10 \times \frac{5}{4} + (5 \times 8) - \left(\frac{25}{8} \times 6 \right) \right) + \left\{ \left(\frac{5}{4} \times 24 \right) - (5 \times 12) + \left(\frac{25}{8} \times 8 \right) \right\} \\ &= -\frac{5}{8} t^2 + \frac{35}{4} t - 5 \end{aligned}$$

$$p(t) = \frac{35}{4} t - \frac{5}{8} t^2 - 5 \quad (\text{Ans.})$$

7(c) For adding new data point, Newton divided difference needs to be used because we won't have to calculate the value of

Date : ____ / ____ / ____

~~new~~ from the beginning.

$$\text{degree} = \text{current degree} + 1 = 2 + 1 = \textcircled{3}$$



◆ Step 1: Lagrange basis

$$L_0(x) = \frac{(x-1.6)(x-1.9)}{0.18}, \quad L_1(x) = \frac{(x-1.3)(x-1.9)}{-0.09}, \quad L_2(x) = \frac{(x-1.3)(x-1.6)}{0.18}$$

◆ Step 2: Derivatives at nodes

$$L_0'(1.3) = -5, \quad L_1'(1.6) = 0, \quad L_2'(1.9) = 5$$

◆ Step 3: Hermite basis functions

$$h_k(x)$$

$$h_0(x) = [1 - 2(x-1.3)(-5)]L_0^2 = (1 + 10(x-1.3))L_0^2$$

$$h_1(x) = [1 - 2(x-1.6)(0)]L_1^2 = L_1^2$$

$$h_2(x) = [1 - 2(x-1.9)(5)]L_2^2 = (1 - 10(x-1.9))L_2^2$$

$$\tilde{h}_k(x)$$

$$\tilde{h}_0(x) = (x-1.3)L_0^2$$

$$\tilde{h}_1(x) = (x-1.6)L_1^2$$

$$\tilde{h}_2(x) = (x-1.9)L_2^2$$

◆ Step 4: Final Hermite polynomial



Ask anything



◆ Step 4: Final Hermite polynomial

$$H(x) = \sum [f(x_k)h_k(x) + f'(x_k)\tilde{h}_k(x)]$$

$$H(x) = 0.620086 h_0(x) - 0.5220232 \tilde{h}_0(x)$$

$$+ 0.4554022 h_1(x) - 0.5698959 \tilde{h}_1(x)$$

$$+ 0.2818186 h_2(x) - 0.5811571 \tilde{h}_2(x)$$

8

◆ Step 5: Put $x = 1.5$

$$f(1.5) \approx 0.5100$$



Ask anything



⑨ 2nd order, degree $\rightarrow 2$, nodes $\rightarrow 3$ | $x_0 = 1$, $x_1 = 2$, $x_2 = 3$

$$P(x) = l_0(x)f(x_0) + l_1(x)f(x_1) + l_2(x)f(x_2) \quad \text{--- (i)}$$

Since $x = 2.1$
so its between
2 and 3. So
close values
are 1, 2, 3

$$l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{(x-2)(x-3)}{2}$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1)(x-3)}{(2-1)(2-3)} = \frac{(x-1)(x-3)}{-1}$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{(x-1)(x-2)}{2}$$

$$\begin{aligned} 2.1 - (-1) &= 3.1 \\ 2.1 - 0 &= 2.1 \\ 2.1 - 1 &= 1.1 \\ 2.1 - 2 &= 0.1 \\ 2.1 - 3 &= -0.9 \end{aligned}$$

Now,

$$P(x) = \left(\frac{(x-2)(x-3)}{2} \times 17 \right) - \left\{ (x-1)(x-3) \times 22.4 \right\} + \left(\frac{(x-1)(x-2)}{2} \times 25.8 \right)$$

$$P(2.1) = \left\{ \frac{(2.1)^2 - 2.1}{2} \times 2.2 \right\} - \left\{ (2.1 - 1) \times 10/6 \right\} + \left\{ \frac{2.1^2 + 2.1}{2} \times 17 \right\}$$

$$= \frac{2173}{100} = 21.73$$

$$P(2.1) = \left\{ \frac{(2.1-2)(2.1-3)}{2} \times 17 \right\} + \left\{ \frac{(2.1-1)(2.1-3)}{1} \times 22.4 \right\} + \left\{ \frac{(2.1-1)(2.1-2)}{2} \times 25.8 \right\}$$

$$= 21.52$$

10. $f(x) = \cos x$

x	$f(x)$
$-\pi/4$	$1/\sqrt{2}$
0	1
$\pi/4$	$1/\sqrt{2}$

$$|f(x) - P_2(x)| = \left| \frac{f^{(3)}(\xi)}{3!} (x-x_0)(x-x_1)(x-x_2) \right|$$

$$= \left| \frac{\sin \xi}{3!} \right| \left| (x+\pi/4)(x-0)(x-\pi/4) \right|$$

$$f(x) = \cos x$$

$$f'(x) = -\sin x$$

$$f''(x) = -\cos x$$

$$f'''(x) = \sin x$$

Now, $\sin x \rightarrow \pi/4$ (max) $= \sin \pi/4 = \frac{1}{\sqrt{2}}$
 $\sin x \rightarrow -\pi/4$ (min) $= \sin(-\pi/4) = -\frac{1}{\sqrt{2}}$

$$\therefore \left| \frac{\sin \xi}{3!} \right| = \frac{1/\sqrt{2}}{3!} = \frac{1}{\sqrt{2} \times 3 \times 2} = 0.11785$$

$$w(x) = (x+\pi/4)(x-0)(x-\pi/4)$$

$$= \left\{ x^2 - (\pi/4)^2 \right\} x$$

$$= x^3 - \frac{(\pi)^2 x}{16}$$

or, $\frac{d}{dx} \left(x^3 - \frac{(\pi)^2 x}{16} \right) = 0$

or,

$$w'(x) = 3x^2 - \frac{\pi^2}{16}$$

$$0 = 3x^2 - \frac{\pi^2}{16}$$

$$x^2 = \frac{\pi^2}{16 \times 3}$$

$$x = \pm \frac{\pi}{4\sqrt{3}}$$

x	$w(x)$
$-\pi/4\sqrt{3}$	0.186
$\pi/4\sqrt{3}$	-0.186

$\therefore$ Upper bound error $= \left| 0.11785 \times 0.186 \right|$

$$= 0.02192$$

11(b)

Chebyshev nodes are considered optimal for interpolation because it prevents Runge phenomenon. Here, the nodes are placed more interpolation points near the endpoints and fewer in the middle because of which the large oscillations of the polynomial near the edges are reduced and makes the approximation more stable.

11(a)

$$f(x) = \sin^2(x/2)$$

Interval $\rightarrow [-1.2, 1.2]$ nodes $\rightarrow 3$ so degree $\rightarrow 2$

x_i	$f(x_i)$
x_0	$-\pi/3$
x_1	0
x_2	$\pi/3$

$$|f(x) - p_2(x)| = \left| \frac{f^{(3)}(\xi)}{3!} (x-x_0)(x-x_1)(x-x_2) \right|$$

$$= \left| \frac{-\sin(\xi)}{3! \times 2} \right| \left| (x+\pi/3)(x-0)(x-\pi/3) \right|$$

$$= \left| \frac{\sin \xi}{3 \times 2 \times 2} \right| \underbrace{\left| (x+\pi/3)(x-\pi/3)x \right|}_{w(x)}$$

Now,

$$\sin(x) \begin{cases} \rightarrow 1.2 (\text{max}) & \sin(1.2) = 0.932 \\ \rightarrow -1.2 (\text{min}) & \sin^2(-1.2) = -0.932 \end{cases}$$

$$\left| \frac{\sin \xi}{12} \right| = \frac{0.318}{12} = 0.0276$$

$$f(x) = \sin^2(x/2)$$

$$f'(x) = (2 \sin x/2) \times 1/2$$

$$= \sin x/2$$

$$f''(x) = \frac{1}{2} \cos x/2$$

$$f'''(x) = \frac{1}{2} \times \frac{1}{2} (-\sin x/2)$$

$$f(x) = \sin^2(x/2) = \frac{1 - \cos x}{2}$$

$$f'(x) = \frac{\sin x}{2}$$

$$f''(x) = \frac{\cos x}{2}$$

$$f'''(x) = \frac{-\sin x}{2}$$

Date : ____ / ____ / ____

$$w(x) = (x + \pi/3)(x - \pi/3)x$$

$$= \{x^2 - (\pi/3)^2\}x$$

$$w(x) = x^3 - \frac{(\pi)^2 x}{9}$$

$$\therefore w'(x) = 3x^2 - \frac{(\pi)^2}{9}$$

$$0 = 3x^2 - \frac{\pi^2}{9}$$

$$3x^2 = \frac{\pi^2}{9}$$

$$x^2 = \frac{\pi^2}{27}$$

$$x = \pm \frac{\pi}{3\sqrt{3}}$$

To get max of a func,

$$\frac{dy}{dx} = 0$$

$$\therefore w'(x) = 0, x \rightarrow \text{max}$$

x	w(x)
-1.2	-0.412
1.2	0.412
$\pi/3\sqrt{3}$	-0.442
$-\pi/3\sqrt{3}$	0.442

maximum

$$\text{Maximum upper bound error} = \left| \frac{0.0776 \times 0.442}{1} \right| = 0.034$$

no. of terms = 3
degree = 3-1 = 2

$x_0 = -0.5, f[x_0] = 1.87$
 $x_1 = 0, f[x_1] = 2.20$
 $x_2 = 0.5, f[x_2] = 2.44$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{2.2 - 1.87}{0 - (-0.5)} = 0.66$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{2.44 - 2.2}{0.5 - 0} = 0.48$$

$$f[x_0, x_1, x_2] = \frac{f[x_2, x_1] - f[x_0, x_1]}{x_2 - x_0} = \frac{0.48 - 0.66}{0.5 - (-0.5)} = -0.18$$

co-efficient

$$P_2(x) = \sum_{k=0}^n \underbrace{n_k(x)}_{\text{newton's polynomial basis}} a_k = n_0(x)a_0 + n_1(x)a_1 + n_2(x)a_2$$

$$\begin{aligned}
 &= 1 \cdot f[x_0] + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] \\
 &= 1.87 + (x - (-0.5))0.66 + (x + 0.5)(x - 0) \times (-0.18) \\
 &= 1.87 + 0.66x + 0.33 + (x^2 + 0.5x) \cdot (-0.18) \\
 &= 1.87 + 0.66x + 0.33 - 0.18x^2 - 0.09x \\
 P_2(x) &= -0.18x^2 + 0.57x + 2.2
 \end{aligned}$$

Date : / /

$$(12)(b) \quad f(x) = \ln(5x+9)$$

Here, $x_0 = 0$

$$a_0 = 2.197$$

$$a_1 = f'(x_0) = \frac{1}{5x+9} \times 5 = \frac{5}{(5 \times 0) + 9} = 5/9$$

$$a_2 = \frac{f''(x_0)}{2!} = \frac{(5x+9) \times 0 - 5(5)}{2! \times (5x+9)^2} = \frac{-25}{(9)^2 \times 2} = \frac{-25}{162}$$

Now,

$$\begin{aligned} P(x) &= f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 \\ &= 2.197 + \frac{5}{9}x + \left(-\frac{25}{162}x\right)^2 \\ &= -0.154x^2 + 0.556x + 2.197 \end{aligned}$$

(12)c) Doesn't match because Newton interpolation uses rounded tabulated values, while Taylor series uses the exact function and its derivatives, leading to differences in co-efficients.

13 (a) yes it will match because both the approaches are just different ways of finding the unique polynomial of degree $\leq n$ that passes through $(n+1)$ nodes.

(13) b

$$P(x) = a_0 + a_1 x + a_2 x^2 \quad \text{--- (i) } \underline{(13) \text{ (c)}}$$

$$P(0) = a_0 + a_1 \cdot 0 + a_2 (0)^2 = 5$$

$$P(3) = a_0 + a_1 \cdot 3 + a_2 \cdot 9 = 9.5$$

$$P(6) = a_0 + a_1 \cdot 6 + a_2 \cdot 36 = 5$$

Now,

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 3 & 9 \\ 1 & 6 & 36 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9.5 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 3 & 9 \\ 1 & 6 & 36 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 9.5 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 3 \\ -1/2 \end{bmatrix}$$

$$\therefore P(x) = 5 + 3x + (-1/2)x^2$$

$$P(x) = -1/2 x^2 + 3x + 5$$

(Ans.)

$$P(x) = l_0(x)P(x_0) + l_1(x)P(x_1) + l_2(x)P(x_2)$$

$$l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-3)(x-6)}{(0-3)(0-6)} = \frac{1}{18}(x-3)(x-6)$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-0)(x-6)}{(3-0)(3-6)} = \frac{x(x-6)}{-9}$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-0)(x-3)}{(6-0)(6-3)} = \frac{x(x-3)}{18}$$

$$P(x) = \left\{ \frac{1}{18}(x-3)(x-6) \times 5 \right\} + \left\{ -\frac{1}{9}x(x-6) \times 9.5 \right\} + \left\{ \frac{1}{18}x(x-3) \times 5 \right\}$$

$$= \frac{5}{18}(x^2 - 9x + 18) - \frac{9.5}{9}(x^2 - 6x) + \frac{5}{18}(x^2 - 3x)$$

$$= x^2 \left(\frac{5}{18} + \frac{5}{18} - \frac{9.5}{9} \right) + x \left(-\frac{5 \times 9}{18} + \frac{6 \times 9.5}{9} - \frac{5 \times 3}{18} \right) + \frac{5 \times 18}{18}$$

$$= -1/2 x^2 + 3x + 5 \quad \text{(Ans.)}$$

(14) a) $p(x) = \frac{8}{5+16x^2}$, $[-6, 6]$, $n=3$, ~~degree = 4~~.

$$\phi_j = \frac{(2j+1)\pi}{2(n+1)}, \quad j = 0, 1, 2, \dots$$

$$\phi_0 = \frac{\{(2 \times 0) + 1\}\pi}{2(3+1)} = \frac{\pi}{8} \quad \phi_2 = \frac{\{(2 \times 2) + 1\}\pi}{2(3+1)} = \frac{5\pi}{8}$$

$$\phi_1 = \frac{\{(2 \times 1) + 1\}\pi}{2(3+1)} = \frac{3\pi}{8} \quad \phi_3 = \frac{\{(2 \times 3) + 1\}\pi}{2(3+1)} = \frac{7\pi}{8}$$

b) $x_j = A \cos \phi_j$ $A = \frac{6 - (-6)}{2} = 6$

$$x_0 = 6 \times \cos\left(\frac{\pi}{8}\right) = 5.543$$

$$x_2 = 6 \times \cos\left(\frac{5\pi}{8}\right) = -2.296$$

$$x_1 = 6 \times \cos\left(\frac{3\pi}{8}\right) = 2.296$$

$$x_3 = 6 \times \cos\left(\frac{7\pi}{8}\right) = -5.543$$

c)
$$L_2(x) = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} = \frac{(x-5.543)(x-2.296)(x+5.543)}{(-2.296-5.543)(-2.296-2.296)(-2.296+5.543)}$$

$$= \frac{(x-5.543)(x-2.296)(x+5.543)}{116.881}$$

15/ (a) we need to find $h_0(x)$, $h_1(x)$, $\hat{h}_0(x)$ and $\hat{h}_1(x)$

$$h_0(x) = \left[1 - 2(x-x_0) \{l'(x_0)\}\right] \times \{l_0(x)\}^2$$

Now,
$$l_0(x) = \frac{x-x_1}{x_0-x_1} = \frac{x-0.2}{0.1-0.2} = \frac{x-0.2}{-0.1}$$

$$l_0'(x) = \frac{1}{-0.1} \quad \therefore l_0'(x_0) = \frac{-1}{0.1}$$

15

Ans. to the Ques. no. (1)

1(a)We have to find $h_0(x)$, $h_1(x)$, $\hat{h}_0(x)$ and $\hat{h}_1(x)$

x	$f(x)$	$f'(x)$
0.1	-0.62050	3.58502
0.2	-0.28340	3.14033

$$h_0(x) = [1 - 2(x - x_0) \{l_0'(x_0)\}] \times \{l_0(x)\}^2 \quad \text{--- (i)}$$

Now,

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - 0.2}{0.1 - 0.2} = \frac{x - 0.2}{-0.1}$$

$$l_0'(x) = -\frac{1}{0.1} \times 1 = -\frac{1}{0.1}$$

$$\therefore l_0'(x_0) = l_0'(0.1) = -\frac{1}{0.1}$$

Putting values of x_0 , $l_0(x)$ and $l_0'(x_0)$ in eq- (i)

$$\begin{aligned} h_0(x) &= [1 - 2(x - 0.1) \times (-\frac{1}{0.1})] \times \left\{ \frac{x - 0.2}{-0.1} \right\}^2 \\ &= [1 - 2\left(-\frac{x}{0.1} + 1\right)] \times \frac{(x - 0.2)^2}{0.01} \end{aligned}$$

Again,

$$h_1(x) = [1 - 2(x - x_1) \{l_1'(x_1)\}] \times \{l_1(x)\}^2 \quad \text{--- (ii)}$$

Now,

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0.1}{0.2 - 0.1} = \frac{x - 0.1}{0.1}$$

$$l_1'(x) = \frac{1}{0.1} \times 1$$

$$\therefore l_1'(x_1) = \frac{1}{0.1}$$

Putting x_1 , $l_1(x)$, $l_1'(x_1)$ in eq—(ii)

$$\begin{aligned} h_1(x) &= \left[1 - 2(x - 0.2) \left\{ \frac{1}{0.1} \right\} \right] \times \left(\frac{x - 0.1}{0.1} \right)^2 \\ &= \left[1 - 2 \left(\frac{x}{0.1} - 2 \right) \right] \times \frac{(x - 0.1)^2}{0.01} \end{aligned}$$

Again,

$$\begin{aligned} h_0(x) &= (x - x_0) \times \{ l_0(x) \}^2 \\ &= (x - 0.1) \times \left(\frac{x - 0.2}{-0.1} \right)^2 = \frac{(x - 0.1)(x - 0.2)^2}{0.01} \end{aligned}$$

$$\begin{aligned} h_1(x) &= (x - x_1) \times \{ l_1(x) \}^2 \\ &= (x - 0.2) \times \left(\frac{x - 0.1}{0.1} \right)^2 = \frac{(x - 0.2)(x - 0.1)^2}{0.01} \end{aligned}$$

15

1(b)

Here, number of nodes $\rightarrow 2$

$$\therefore \text{degree} = (2n+1) = (2 \times 2) + 1 = 5$$

$$P_5(x) = h_0(x) f(x_0) + \hat{h}_0(x) f'(x_0) + h_1(x) f(x_1) + \hat{h}_1(x) f'(x_1)$$

$$= \left[1 + \frac{2x}{0.1} - 2 \right] \times \frac{(x-0.2)^2}{0.01} \times (-0.62050) + \frac{(x-0.1)(x-0.2)^2}{0.01} \times$$

$$3.58502 + \left[1 - \frac{2x}{0.1} + 4 \right] \times \frac{(x-0.1)^2}{0.01} \times (-0.28340) +$$

$$\frac{(x-0.2)(x-0.1)^2}{0.01} \times (3.14033)$$

$$= \left[\frac{2x}{0.1} - 1 \right] \times (-62.05 \times (x-0.2)^2) + 358.502 (x-0.1)$$

$$(x-0.2)^2 + \left[5 - \frac{2x}{0.1} \right] \times (-28.34 (x-0.1)^2) +$$

$$(x-0.2)(x-0.1)^2 \times 314.033$$

$$P_5(0.15) = \left(\frac{2 \times 0.15}{0.1} - 1 \right) \times \left\{ -62.05 \times (0.15-0.2)^2 \right\} +$$

$$358.502 (0.15-0.1) (0.15-0.2)^2 + \left(5 - \frac{2 \times 0.15}{0.1} \right) \times$$

$$\left\{ -28.34 \times (0.15-0.1)^2 \right\} + \left\{ (0.15-0.2) (0.15-0.1)^2 \times 314.033 \right\}$$

$$= 2 \times (-0.155125) + 0.04481275 + 2 \times (-0.07085) + (-0.039254125)$$

$$P_5(x) = -0.4463913$$